

Quiz Covid

$$2 \int_0^{\frac{\sqrt{3}}{2}} \int_{\frac{1}{2}}^{\sqrt{1-x^2}} x+y \, dy \, dx = 2 \int_0^{\frac{\sqrt{3}}{2}} \left| xy + \frac{y^2}{2} \right|_{\frac{1}{2}}^{\sqrt{1-x^2}} dx =$$

$$= 2 \int_0^{\frac{\sqrt{3}}{2}} \frac{1-x^2}{2} + x\sqrt{1-x^2} - \frac{1}{8} - \frac{1}{2}x \, dx = \int_0^{\frac{\sqrt{3}}{2}} 1-x^2 + 2x\sqrt{1-x^2} - \frac{1}{8} - x \, dx =$$

$$= \left| \frac{3x}{8} - \frac{x^3}{3} + \frac{2}{3}(1-x^2)^{\frac{3}{2}} - \frac{x^2}{2} \right|_0^{\frac{\sqrt{3}}{2}} = \frac{3\sqrt{3}}{8} - \frac{3\sqrt{3}}{24} + \frac{1}{12} + \frac{3}{8} - \frac{2}{3} = \frac{\sqrt{3}}{4} \quad ;$$

✓ 2

$$f = x^2 + y^2 \quad f_x = 2x \quad f_y = 2y$$

$$\frac{\partial F_2}{\partial x} = 2 + \frac{-x^2 - y^2 + 2x^2}{(x^2 + y^2)^2} = 2 + \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial F_1}{\partial y} = \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \quad \text{not } = 2$$

$$\int_0^{2\pi} \int_0^1 2p \, dp \, d\sigma = \int_0^{2\pi} |p^2|_0^1 \, d\sigma = 2\pi$$

✓ 1

$$\begin{cases} t^2 + 1 = 5 \\ 2t = 4 \end{cases} \begin{cases} t = 2 \\ j = (2t_0, 2) = (4, 2) \end{cases} \quad 4(x-5) + 2(y-4) = 0$$

$$-2x + 10 + 4y - 16 = 0 \quad 4y = 2x + 6 \quad y = \frac{1}{2}x + \frac{3}{2}$$

$$j = (2t, 3) = (4, 3)$$

$$-3(x-5) + 4(y-6) = 0$$

$$-3x + 15 + 4y - 24 = 0$$

$$4y = 3x + 9$$

$$y = \frac{3}{4}x + \frac{9}{4}$$

Exemple 2

$$1) \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + 2y^4} = \frac{y^4}{3y^4} = \frac{1}{3} \quad \frac{x^3 \cdot m^2}{x^2 + 2x^4 m^4} = 0$$

$$2) rot = 2x - 3y^2$$

$$\int_0^{2\pi} \int_0^1 (2p^2 \cos \theta - 3p^3 \sin^2 \theta) dp d\theta = \int_0^{2\pi} \cos \theta \left| \frac{2}{3} p^3 \right|_0^1 - \sin^2 \theta \left| \frac{3}{4} p^4 \right|_0^1 d\theta =$$

$$= \int_0^{2\pi} \left(\frac{2}{3} \cos \theta - \frac{3}{4} \sin^2 \theta \right) d\theta = \frac{3}{4} \pi$$

$$x^2 + y^2 + z^2 = 1$$

ESAME 1

P(4,2)

$$4(y-4) = 2(x-5)$$

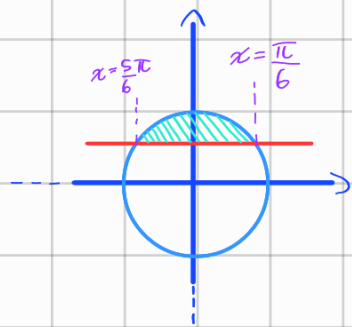
$$4y = 2x + 6$$

$$y = \frac{1}{2}x + \frac{3}{2}$$

2) rot = 2 curva chiusa $\vec{r} = (\cos t, \sin t)$

$$2 \int_0^{2\pi} \int_0^1 p = 2\pi$$

es 3



$$\left\{ p \cdot \sin \theta > \frac{1}{2}, \frac{\pi}{6} \leq \theta \leq \frac{5\pi}{6} \right\}$$

$$\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \int_{\frac{1}{2\sin\theta}}^1 p^2 \cos \theta + p^2 \sin \theta = \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (\cos \theta + \sin \theta) \cdot \left(\frac{p^3}{3} \right) \Big|_{\frac{1}{2\sin\theta}}^1 =$$

$$= \frac{1}{3} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \cos \theta + \sin \theta - \frac{1}{8} \left(\frac{\cos \theta + 1}{\sin^3 \theta} + \frac{1}{\sin^2 \theta} \right) =$$

$$= \frac{1}{3} \left| \sin \theta - \cos \theta + \frac{3}{8 \sin^2 \theta} + \frac{\cot(\theta)}{8} \right|_{\frac{\pi}{6}}^{\frac{5\pi}{6}} =$$

$$= \frac{1}{3} \left(\frac{1+\sqrt{3}}{2} - \frac{\sqrt{3}}{8} - \left(\frac{1+\sqrt{3}}{2} - \frac{\sqrt{3}}{8} \right) \right) = \frac{\sqrt{3}}{4}$$

es 4

$$f = p^2 + \frac{1}{2p^2}$$

$$f' = 2p - \frac{1}{p^3}$$

$$p^4 = \frac{1}{2} \quad p = \frac{1}{\sqrt[4]{2}}$$

ESAME 2

$$① \frac{p^3 \cdot \cos \theta \cdot \sin^2 \theta}{p^2 \cdot \cos^2 \theta + 2p^4 \cdot \sin^4 \theta} = 0 \quad \frac{y^4}{3y^4} = \frac{1}{3} \quad \cancel{\neq}$$

$$② \quad r_{\text{ot}} = 2x - 3y^2 \quad \gamma = (\cos t, \sin t) \quad I = [0, 2\pi]$$

$$\int_0^{2\pi} \int_0^1 (2x - 3y^2) dx dy = \int_0^{2\pi} \int_0^1 (2p^2 \cos \theta - 3p^3 \sin^2 \theta) =$$

$$= \int_0^{2\pi} \left(\frac{2 \cos \theta}{3} - \frac{3}{4} \sin^2 \theta \right) d\theta = \frac{-3}{4} \left| \frac{1}{2} \theta - \frac{\sin(2\theta)}{4} \right|_0^{2\pi} =$$

$$= -\frac{3}{4} (\pi) = -\frac{3}{4} \pi$$

$$\left\{ \frac{1}{2} \leq z \leq 1, \frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}, 0 \leq \varphi \leq 2\pi \right\}$$

$$2\pi \cdot \left(\frac{2}{3} \cdot \frac{1}{2} \right) = \frac{2}{3} \pi$$

$$\frac{1}{2} - \frac{1}{3} = \frac{6-2}{6} = \frac{2}{3}$$